Relaxed Randomized Gossip on the Complete Graph: Exact Covariance Spectrum and Sharp Consensus Law

Route-A source-local certificate HCS-C333

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Abstract

For uniform pairwise relaxed averaging on the complete graph, we derive the exact first moment and sharp mean-square disagreement law at every finite time. The entire second-moment transfer on $\mathrm{Sym}^2(\mathbf{1}^\perp)$ splits into three explicit orthogonal blocks; all eigenvalues, multiplicities, projectors, and matrix-valued iterates are closed. This yields the exact statistical covariance, a finite-time tail bound, almost-sure consensus for $0 < \eta < 1$, and a complete treatment of $N = 1, 2, 3$ and $\eta = 0, 1$. In particular, $\eta = 1$ is a random-transposition boundary and not a consensus regime.

1 Frozen stochastic owner

Fix $N \geq 2$ and $0 \leq \eta \leq 1$. Independently at every integer time, choose an unordered pair $\{i, j\}$ uniformly from the edges of K_N and set

$$x_{t+1} = W_{ij} x_t, \qquad W_{ij} = I - \eta d_{ij} d_{ij}^\mathsf{T}, \qquad d_{ij} = e_i - e_j. \tag{1}$$

Thus the selected entries become $(1-\eta)x_i + \eta x_j$ and $\eta x_i + (1-\eta)x_j$. The endpoint $\eta = 1/2$ is ordinary pairwise averaging, while $\eta = 1$ merely swaps the two entries. The initial vector is deterministic.

Put

$$P = I - \frac{1}{N}\mathbf{1}\mathbf{1}^\mathsf{T}, \qquad \bar{x} = \frac{1}{N}\mathbf{1}^\mathsf{T} x_0, \qquad y_t = P x_t = x_t - \bar{x}\mathbf{1}.$$

Every W_{ij} is symmetric, fixes $\mathbf{1}$, and commutes with P. The basic complete-graph identity is

$$\sum_{i<j} d_{ij} d_{ij}^\mathsf{T} = NP. \tag{2}$$

Theorem 1 (Mean and sharp energy law)**.** *The sample mean is invariant on every realization. With*

$$\mu = 1 - \frac{2\eta}{N-1}, \qquad \rho = 1 - \frac{4\eta(1-\eta)}{N-1}, \tag{3}$$

one has, for every $t \geq 0$,

$$\mathbb{E} y_t = \mu^t y_0, \qquad \mathbb{E}\|y_t\|^2 = \rho^t \|y_0\|^2. \tag{4}$$

Proof. Equation (1) fixes $\mathbf{1}^\mathsf{T} x$ pathwise. Since K_N has $N(N-1)/2$ edges, averaging (1) and using (2) gives

$$\mathbb{E}W = I - \frac{2\eta}{N-1} P,$$

which acts by μ on $\mathbf{1}^\perp$. Independence of the new edge from the past proves the first identity by iteration. Since $(d_{ij} d_{ij}^\mathsf{T})^2 = 2 d_{ij} d_{ij}^\mathsf{T}$,

$$W_{ij}^2 = I - 2\eta(1-\eta) d_{ij} d_{ij}^\mathsf{T}.$$

Conditioning on y_t, applying (2), and iterating gives the second identity. It is an equality for every initial vector, hence its rate is sharp whenever $y_0 \neq 0$. $\square$

The phrase *sharp energy owner* records the content added in the original paper round: no spectral conclusion is needed for Theorem 1.

2 Three invariant covariance blocks

Let

$$\mathscr{H}_N = \{A \in \mathbb{R}^{N\times N} : A = A^\top,\ PAP = A\}$$

with Frobenius inner product. For $N \geq 3$ define

$$\begin{aligned}\mathscr{V}_0 &= \operatorname{span}\{P\},\\ \mathscr{V}_1 &= \{P\operatorname{diag}(u)P : \mathbf{1}^\top u = 0\},\\ \mathscr{V}_2 &= \{A \in \mathscr{H}_N : \operatorname{diag} A = 0\}.\end{aligned}$$

For $N = 3$, $\mathscr{V}_2 = \{0\}$. For $N = 2$, only $\mathscr{V}_0 = \mathscr{H}_2$ remains.

Lemma 2 (Explicit orthogonal projectors)**.** *For $N \geq 3$ and $A \in \mathscr{H}_N$, set*

$$\begin{aligned}\Pi_0 A &= \frac{\operatorname{tr} A}{N-1}P, \qquad R = A - \Pi_0 A,\\ u &= \frac{N}{N-2}\operatorname{diag} R, \qquad \Pi_1 A = P\operatorname{diag}(u)P, \qquad \Pi_2 A = A - \Pi_0 A - \Pi_1 A.\end{aligned} \tag{5}$$

Then Π_r are the orthogonal projectors onto $\mathscr{V}_r$, and

$$\dim \mathscr{V}_0 = 1, \qquad \dim \mathscr{V}_1 = N-1, \qquad \dim \mathscr{V}_2 = \frac{N(N-3)}{2}. \tag{6}$$

Proof. The first residual has trace zero, so $\mathbf{1}^\top u = 0$. For such u,

$$\operatorname{diag}(P\operatorname{diag}(u)P) = \frac{N-2}{N}u.$$

Hence $\Pi_1 A$ has the same diagonal as R, and $\Pi_2 A$ has zero diagonal. Centered matrices with zero diagonal are Frobenius orthogonal to both P and every centered diagonal pattern. This proves orthogonality and the projector formulas. The map $u \mapsto P\operatorname{diag}(u)P$ is injective on $\mathbf{1}^\perp$ for $N > 2$. Subtracting dimensions from $\dim \mathscr{H}_N = N(N-1)/2$ proves (6). □

Define the second-moment transfer

$$\mathcal{T}_\eta(A) = \frac{2}{N(N-1)}\sum_{i<j} W_{ij} A W_{ij}. \tag{7}$$

Theorem 3 (Complete second-moment spectrum)**.** *The three spaces in Lemma 2 are orthogonal invariant blocks of $\mathcal{T}_\eta$ for every $0 \leq \eta \leq 1$. The restrictions to the present nonzero blocks are scalar, with respective values*

$$\lambda_0 = 1 - \frac{4\eta(1-\eta)}{N-1}, \tag{8}$$

$$\lambda_1 = 1 - \frac{4\eta - 2\eta^2}{N-1}, \tag{9}$$

$$\lambda_2 = 1 - \frac{4\eta}{N-1} + \frac{4\eta^2}{N(N-1)}. \tag{10}$$

Absent low-dimensional blocks are omitted. If $\eta > 0$, the values attached to the present blocks are pairwise distinct, so those blocks are precisely the corresponding full eigenspaces. If $\eta = 0$, then $\mathcal{T}_0 = I$: the only eigenvalue is one, its eigenspace is all of $\mathscr{H}_N$, and its multiplicity is $N(N-1)/2$.

Proof. For centered A, expansion of (7) gives

$$\mathcal{T}_\eta(A) = (1 - \frac{4\eta}{N-1})A + \frac{2\eta^2}{N(N-1)}\mathcal{S}(A), \quad \mathcal{S}(A) = \sum_{i<j}(d_{ij}^\top A d_{ij})d_{ij}d_{ij}^\top. \tag{11}$$

If $A = P$, every scalar coefficient in $\mathcal{S}$ is two, so $\mathcal{S}(P) = 2NP$. If $A = P\operatorname{diag}(u)P$ with $\mathbf{1}^\top u = 0$, then $d_{ij}^\top A d_{ij} = u_i + u_j$. The off-diagonal and diagonal entries give $\mathcal{S}(A) = NA$. Finally, for $A \in \mathscr{V}_2$, $d_{ij}^\top A d_{ij} = -2a_{ij}$; its zero row sums then give $\mathcal{S}(A) = 2A$. Substitution in (11) yields the three displayed eigenvalues. For $\eta > 0$,

$$\lambda_0 - \lambda_1 = \frac{2\eta^2}{N-1} > 0, \qquad \lambda_1 - \lambda_2 = \frac{2\eta^2(N-2)}{N(N-1)} > 0,$$

where the second comparison is used only when $\mathscr{V}_2$ is present. Together with the orthogonal decomposition, this proves the full-eigenspace statement. At $\eta = 0$ every $W_{ij} = I$, which proves the asserted merger. □

Theorem 4 (Exact second moment)**.** *Let $A_0 = y_0 y_0^\mathsf{T}$ and $M_t = \mathbb{E}[y_t y_t^\mathsf{T}]$. For $N \geq 3$,*

$$M_t = \lambda_0^t \Pi_0 A_0 + \lambda_1^t \Pi_1 A_0 + \lambda_2^t \Pi_2 A_0, \tag{12}$$

where the last term is absent at $N = 3$. At $N = 2$, $M_t = \lambda_0^t A_0$.

Proof. The newly selected edge is independent of the past, hence $M_{t+1} = \mathcal{T}_\eta(M_t)$. Apply Theorem 3 and iterate. Taking traces recovers Theorem 1, because the two trace-free blocks do not contribute. □

3 Covariance and probabilistic closure

The disagreement second moment M_t is not, in general, the statistical covariance: $\mathbb{E}y_t = \mu^t y_0$ need not vanish. The exact covariance is therefore

$$\mathrm{Cov}(x_t) = M_t - \mu^{2t} y_0 y_0^\mathsf{T}. \tag{13}$$

Both terms annihilate the consensus direction.

Theorem 5 (Tail bound and almost-sure consensus)**.** *If $N \geq 2$, $0 < \eta < 1$, $\varepsilon > 0$, and $y_0 \neq 0$, then*

$$\Pr\{\|y_t\| \geq \varepsilon \|y_0\|\} \leq \frac{\rho^t}{\varepsilon^2}. \tag{14}$$

For every initial vector, including $y_0 = 0$, one has $x_t \to \bar{x}\mathbf{1}$ almost surely and in mean square.

Proof. Here $0 \leq \rho < 1$. Markov's inequality and Theorem 1 give (14); division by $\|y_0\|$ is legitimate under the stated hypothesis. If $y_0 = 0$, every update fixes the initial consensus vector. Otherwise, for each positive rational ε, the probabilities in (14) are summable in t. Borel–Cantelli, followed by a countable intersection, gives $\|y_t\| \to 0$ almost surely. Mean-square convergence is already the exact energy identity. □

4 Complete boundary atlas

For $N = 1$ there is no edge to sample; by definition the process is static. For $N = 2$ the disagreement line is one-dimensional and its difference is multiplied by $1 - 2\eta$ at every step. Hence $\eta = 1/2$ reaches consensus in one update. For $N = 3$, the third block has dimension zero, so inserting a λ_2 multiplicity would be spurious.

At $\eta = 0$, all W_{ij} and $\mathcal{T}_0$ are the identity. The three invariant summands therefore merge into the eigenvalue-one eigenspace $\mathscr{H}_N$, of total multiplicity $N(N-1)/2$; they must not be called three distinct eigenspaces at this endpoint. At $\eta = 1$, each W_{ij} is the transposition of coordinates i and j, and $\rho = 1$; non-consensus data cannot converge to consensus because their disagreement norm is constant. This is the precise random-transposition boundary, not an extension of the interior convergence theorem. Constant initial vectors are fixed for every parameter. Values $\eta \notin [0, 1]$, nonuniform edge laws, and noncomplete graphs lie outside the frozen owner.

5 Evidence, collisions, and Route-A boundary

The exact certificate contains 56 spectral rows, six projector receipts, 48 exhaustive-word rows, and 2,966 scalar leaves. It enumerates all 4,242 edge words in its declared small-time grid. A producer-independent checker performs 1,392 checks, while the SymPy lane closes 350 exact identities. Two isolated producer runs are byte-identical, and 140 repaired-hash, parser, theorem, boundary, and evaluator attacks are rejected. The finite grid is regression evidence; the preceding theorems are analytic for every declared N and η.

The closest registered systems are C183, which owns the permutation-valued random-transposition chain; C203, deterministic signed-Laplacian consensus; C312, state-dependent Hegselmann–Krause averaging; and C322, Kac's continuous-angle collision operator. None owns the interior relaxed random product and its full second-moment decomposition.

The phrase *probability boundary and route firewall* records the final revision. Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the Route-A tuple is

$$(\text{A0\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FORMAL\_HINT}).$$

Pair maps and the moment transfer are symmetric on natural source spaces, but this is only a formal lift hint. Random pair words are not deterministic isolated prime-labelled cycles, and interaction count is not a logarithmic prime clock. There is no orbit zeta, target Fredholm determinant, functional equation, Weil compression, target divisor, target-zero match, or natural same-clock unitary quantization. Route A is rejected and Route B remains locked. No target arithmetic local data, Euler factors, root number, automorphy, or Hilbert–Polya operator is claimed.

AI use. A generative language model assisted proof organization, code scaffolding, and manuscript drafting. The displayed derivations, independent exact checker, symbolic lane, hostile tests, and deterministic artifacts define the audit record.

References

[1] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, "Randomized gossip algorithms," *IEEE Transactions on Information Theory* 52 (2006), 2508–2530. DOI: 10.1109/TIT.2006.874516.

The source establishes the randomized-gossip owner and network setting. The complete-graph formulas above are derived directly for the frozen model; no literature-priority claim is made for them.